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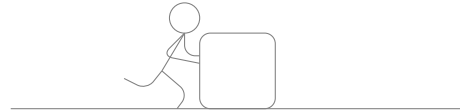
Free-Body Diagrams & Single-Body Force Problems

1. For each of the sketches below, identify all the forces applied on all objects and draw a free body diagram. Then come up with an expression for the net force.

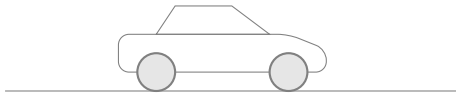
(a) Lamp hanging from a chain



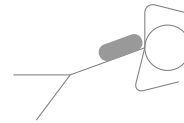
(d) A box being pushed forward on the ground (constant speed)



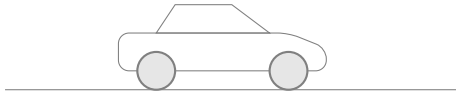
(b) A car moving at a constant speed.



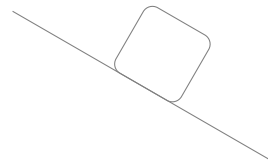
(e) A skydiver before opening her parachute



(c) A car accelerating



(f) Object sliding down an inclined plane.



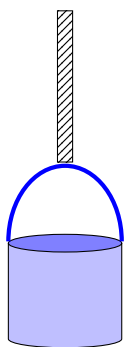
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2. Neal is stealing a refrigerator from the bank. The mass of the refrigerator is 173 kg and its kinetic friction has a magnitude of 254 N. How hard must he push it forward in order to accelerate it at 1.27 m/s^2 ?

3. Joe pulls up on a rope attached to a 5.5-kg bucket. The bucket accelerates at 2.1 m/s^2 . With what force did Joe pull?



4. You ($m = 46 \text{ kg}$) are standing on an elevator on the top floor of a building. The elevator begins to go down towards the ground floor and does so by accelerating downward at 2.7 m/s^2 . What is the normal force acting on you?

